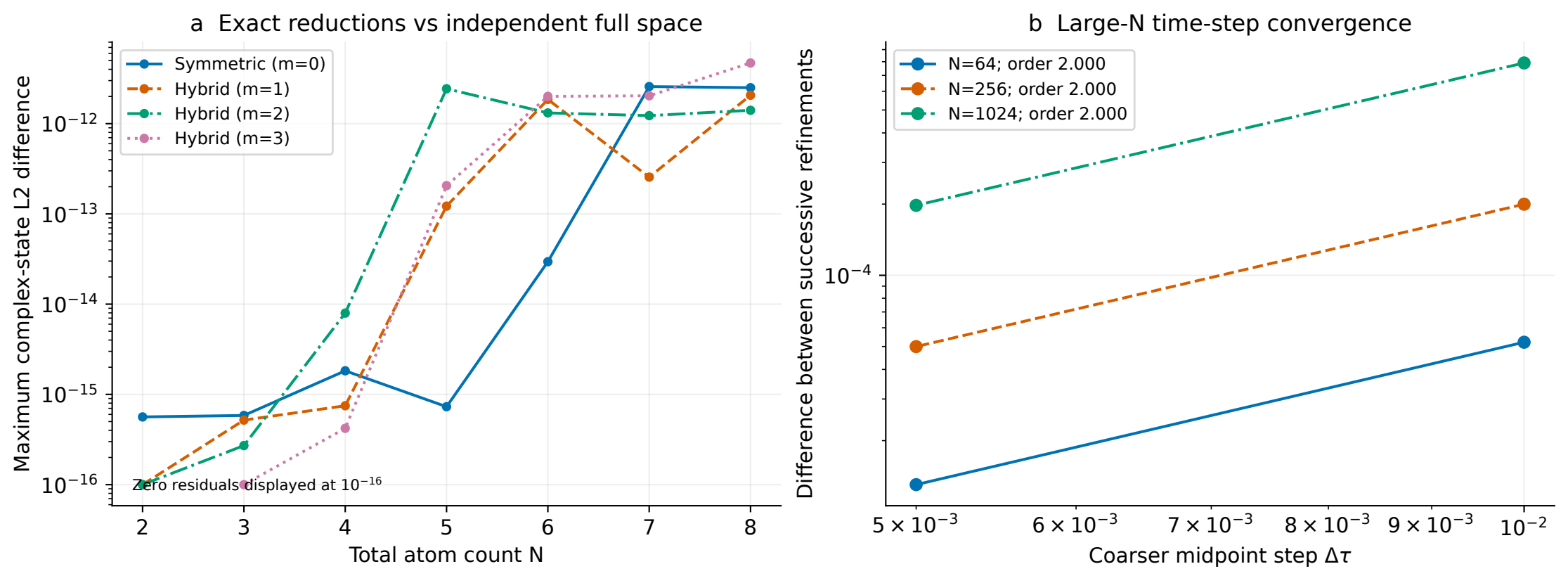
a Exact reductions vs independent full space
Symmetric (m=0)
Hybrid (m=1)
Hybrid (m=2)
Hybrid (m=3)
Maximum complex-state L2 difference
10^{-12}
10^{-13}
10^{-14}
10^{-15}
10^{-16}
Zero residuals displayed at 10^{-16}
2
3
4
5
6
7
8
Total atom count N
b Large-N time-step convergence
N=64; order 2.000
N=256; order 2.000
N=1024; order 2.000
Difference between successive refinements
10^{-4}
5×10^{-3}
6×10^{-3}
7×10^{-3}
8×10^{-3}
9×10^{-3}
10^{-2}
Coarser midpoint step $\Delta\tau$